What is your current role? What do you do?
I am a senior developer.

And how long have you been working with software development?
For at least 5 years.

And which AI tools do you use in your work for software development?
For software development, specifically, I use Cursor AI.

And how often do you use this tool?
Every day.

And in which types of activity do you usually use it?
Ah, documentation and software development itself.

What type of documentation?
Okay, there are some types of documentation, depending on the activity I am performing. Let's suppose it is a refactoring activity. So, first we document what needs to be refactored, and for that we use AI to help, to catalog what is currently working, build an architecture diagram of how it currently works, and propose a new solution on top of that. And all of this will generate documentation that later we will publish on Confluence. It will be available for everyone to see.

But is this Markdown documentation?
Yes, yes, Markdown. In the case of Confluence, it does not accept markdown, so we kind of have to manually convert what was in markdown into Confluence documentation, but originally in markdown.

And can you describe a recent case in which you used AI to perform a development activity, from beginning to end?
So, I will take an example of refactoring that we did. Ah, we had a home screen, remembering that, currently, [company name], it is a consulting company and I am allocated to a client. So, the client I am allocated to is [client name]. It is a company that works with TV, so I develop for Android TV. And in their application, we have the home screen and this home screen will have a new feature, which I cannot say exactly what it would be, but on this new screen, on the old screen, we would only have one type of content. We only had one type of content. And we started to implement more types of content that will be displayed. The problem is that in the architecture that existed before, it was made to support only one type of content, which is TV content. Not third-party content. So, the home screen is composed of three main parts, right? The part where the video player is, the recommendations part, and the part with our favorites that stays up there. And each of these parts loads independently. But when we had only one type of content, we knew exactly how to obtain this type of content. Since we started to have different types of content, each type of content needed, started to use a different service, and the old architecture was not adequate to have more than one service, it was rigid for only one type of service. So, with the help of AI, we planned how this refactoring would be done, to divide the services into layers, so we added new layers of abstraction so that we could have different types of content coming from different services.

After we built the initial architecture, we created documentation, documentation on how the migration would be done part by part, which new classes would be added, which new services would be added, the new models that would be added. Ah, we also added an asynchronicity layer, because different services could be obtained asynchronously and before they were being done synchronously. Ah, the AI identified how to do this and we were able to document it. Of course, after the documentation is ready, we have to validate it, because sometimes the AI hallucinates, sometimes it says that something can be done when actually it cannot be done. So, we go through that documentation, we make a draft there, we start prototyping this implementation, see what works and what does not, feed the AI again telling it the things that can be done and that cannot be done, and then we have a final documentation. With this final documentation, from that, we use AI to, anyway, code what we sat down and said: "Well, this is what we are going to do." And we let the AI work. And after it finishes its work, we have to do a revalidation, right? It is very important, for example, that sometimes the AI does things, but it does not check whether the guidelines are being followed, sometimes it does not check whether the code compiles, so it just does it and says trust me there, right? So it is important that we give it some instructions, for example, ah, one thing, one problem I faced a lot, is in the matter of logging, of logging things in the console, that sometimes, after it is ready, we need to check, do manual validation, it is very good to have logs, right? And Android has its standard log, right, which is that log.d. But in our project we do not use log.d, we use JF4, something like that. Which is another logging system. So, the AI wants to use its standard, but we need to state to it, “use this logging standard.” Ah, another thing, keep the dependency injection pattern. So, sometimes it wanted to use a static class and that is not supposed to be done. So, we need to come and tell it, look, use the dependency injection issue that is configured in such-and-such file, do it in such-and-such way. We needed to give these small instructions. And sometimes it just threw code that did not compile. So, we also needed to tell it that at the end of the implementation, try to compile the code, check what went wrong, and fix it. So, do this whole process. Then after leaving the AI running there for some time, then comes the part of doing the manual validation and the manual fine adjustments, because sometimes the AI does not understand the whole context, because it is an extremely large and complex application, it gets very lost, so we try to reduce the scope there only to that home screen so it does not keep looking in other places that have nothing to do with what we need. After all these validations, we delivered the functionality. Of course, afterwards we still had to go back and redo some more adjustments, because as it was a very large refactoring, it modified many, many, many files. So in the code review part, for example, I was the one who did this part, but then, at code review time, I probably let some things pass that I had not seen, like, some log that ended up passing, some import that should not be there. So, in the code review part we were able to prune these small errors that I ended up letting pass in the first round of the AI.

Is this code review necessary because you were the one who did it? You asked all the instructions to the AI. So, you could probably be biased and asked another person to do it?
Exactly. My mind was biased. I had already done it, after doing it, I did my first review. But of course, right? It is a lot of things. And sometimes, just glancing over it there, I will end up letting some things slip. So it is important to have this review from other developers, because they will have a fresh mind, they will be there, they will catch it faster. Oops, this here does not make sense. Oops, this was missing here. Hey, why is it this way here? Why couldn't it be another way? It is important to have this review afterwards.

So basically, how many steps do you follow and which documentations do you generate in each AI step? This first documentation you generate, what is it? Do you call it a PRD? Would it be that or not?
No, this is what we call a Spike. The Spike is usually a story. A story. Then each story has, there are types of story. There is task story. There is task, there is story and there is Spike. The Spike is a story, but it is focused on investigation. So, let's suppose like this, man, we are going to implement a new feature. So, PO already brought me there an initial documentation, the designer gave me another documentation.

But these documentations they are passing to you are also Markdown, made by AI or not?
They are documents made by them, at least I know that the ones the POs send are reviewed by AI. But they give me the requirements there. So I take the requirements and from the

requirements, I will do the spike. What is the spike? I will investigate how I am going to implement this requirement within my existing code. So, man, I need to see how I am going to create this new button. Like, there is a green button and I need to implement a red button on a screen. So, what would the spike be? The spike, I will investigate how I am going to implement this green button on this screen. This green button that I am going to implement on this screen, will I need to create a new component? Can I reuse an existing component? If I add this button to my screen, will it break my layout? This new button that I am going to add, what is its behavior? Should it keep the same behavior as the red button that I already had on my screen? Will it have a new behavior? Will it break some user flow? So, this is the investigative part that I am going to do. I will investigate in the code, see what the difficulties will be to add this new component, what will be new, whether I will be able to reuse some existing component, if I add this button here, I discovered that actually it was not as simple as I imagined. So, it is discovering how to do it, why to do it, and what impediments I will probably find. Man, maybe this new button that I am going to add has a dependency that I did not even expect. Man, because the click on this button, actually, it has to create a new service. Man, so this is something that I did not have, that I did not expect. Let's suppose, there were two buttons and this green button called a new API. This is something I had not captured in the requirements, for example. Then I realized that, man, this API that it is asking me to call does not exist in the code. So, I will have to add one more layer, one more service to call this API. So, all of this will be documented in this Spike story. So, I will generate a document, I will share it with the team and from this Spike I can create new stories. It may be only one story, saying: "Ah, OK." At the end it will be there: Ah, it may be that this spike generates one or more stories. At the end, either it will generate a story saying: "Ah, it is just to implement it like this, like that, done." Or it will say: "No, to implement this, first I have to refactor a part of what is implemented, this becomes a refactoring story." I also need another story to do the, to create the new services and finally a story to implement all of this. That is, this spike that I did, this investigation that I did, this documentation that I generated, it can create new stories or even new documentations.

And then from this spike you discuss with the team whether these stories really are, or whether you will create more. Or do you follow what it is recommending?
Yes, we discuss with the team. Then the next process is basically this. Generate documentation, analyze documentation, and from that we start coding. And in the coding stage there are still some things, okay. We have the initial activity, right? Getting to the implementation part, we do not simply ask the AI to do things. The first step is that we create a plan for the AI to do it, that is, okay, you come implement this button. I will not simply tell the AI to implement this button, I will say: create a plan of how you will implement this button. So, it will describe to me exactly what it is going to do, it will say the steps it will take before implementing. I will validate whether this plan makes sense. So, I will look if it makes sense. Let's suppose: "Ah, I want to implement this green button." Then, it puts there that it will say a blue color. Since I asked it to implement the green button. That is, in the planning part of what it is going to do, I can already catch it before it even creates the code. So, without having to touch the code yet, I have a plan document where I can see its reasoning, in a way, of what it is going to do. I validated that the plan is all right. Then, I particularly do it this way. I ask it to create a progress document. In this progress document, it will put all the step by step of what it has to do and I ask it, while it is implementing, to mark what it did in the progress document. This way, it keeps a memory of what it did until the implementation is finished. And at the end of the implementation, it has to compile the code and make sure it is working. It cannot give me code that does not compile.

And this document, was it you who gave the instructions for what it should have, let's say, its template, or does it have the template and you just ask it to use this progress template?
The progress template, no, I do not have it, I do not have it. I just ask it to do it. I usually use a very simple prompt. Create a document called progress.md and put the step-by-step and at each iteration of what you are doing, mark it as completed.

For the stage of your planning, do you pass this spike that it created and then based on it, there is a story there and then you say: Ah, make the planning for this story. Is that it? Or not? You do not pass anything when you are going to code.
No, the spike itself, it is already a separate thing. So, after the Spike, it will generate a new story. Obligatoriamente, it will generate a new story. Either it will generate a new story, or it will say: "It cannot be done." I understand. Which is very rare, but usually it will generate a new story. And from this new story, then it will be well described, it is expected that it is well described. We will take it and pass part of the story and what I read in the Spike that I think is necessary, I add to the prompt. I ask it, you will do this, this, this. Yes, I also have some documents with some project standards. And there are some project standards. For example, it is not to use Android log.D, it is to use JF4. So, there are some standards, ah, use Dagger dependency injection, do other things. So, I pass these standards to it in the prompt and say what it has to do and ask it to generate the plan.

And this is for a more, let's say, large context, right? Building a feature, when you are going to do something more basic, do you follow another process?
Yes, for example, in this case, when something to do is very large, Cursor itself has a planning function. But when it is something very large, I think this planning function does not work so well. At least in my opinion, it does not work so well. So I ask the AI to generate its own document. Usually, there is another thing, usually, in Cursor we have several models that we can use. To generate the document I usually use the model that is those models that think more, they spend more tokens, but they think more before giving me an answer and usually they generate a more detailed document. Than I simply use a simple model, that flat model that just answers you.

And for simpler tasks, how do you do your process?
Okay, let's suppose I have a very simple task. Yes, I have to implement a button, for example. Ah, it is just making a button, there is not much secret, but we have a design in Figma, for example. For this, we have a Figma MCP server, that we can get this design and throw it directly into the AI, and it can generate my component for me without my needing to make many adjustments. I even generated one recently, it turned out well. Yes, but of course, there are some details, some nuances maybe that just from the component it will not be able to capture. For example, I asked it to generate a component but I did not give it much context. So, it only generated the component the way it thought it should be generated. It generated correctly, but it did not have much functionality. For example, it generated, I had, right below the component, there was a little metadata bar. But this metadata bar, it was supposed to be a list. Which the AI generates only as if it were text. That is, it did not catch this nuance that actually it was a list. So, I had to ask it to modify it to generate a list and not simply a text bar. But then in this mode I did not ask for planning, I did not ask for anything, I just said: "Do it, the model is here, it is to do it like this, follow such-and-such convention and do it." I only asked.

And when you go, for example, you asked it to make this component? And in this case it did it wrong, then, when you want to adjust it, do you go directly into the code or do you keep asking it and then it adjusts according to what you are asking?
Yes, if it is something a little complex, I ask it to adjust. If it is something very simple, I adjust it manually. For example, in this case of this component, there was something complex and there was something simple. What was the complex thing? It was this metadata list, because this metadata list by itself ended up becoming a new component. So, this new component that I needed to generate, it had a nuance of how it should behave. So, I asked the AI for this, I explained to it how it should be done and it did it. Now another component that was very simple, which was simply the color of the progress bar that was wrong. Then this one, I went there in Figma, saw what the color was and added it. I did not need to use AI for that.

And has the use of AI changed the way you work? If so, how?
Yes, it changed a lot because before I had a concern with understanding the code, seeing what I needed to do and starting to type, right? Starting to type, or copy and paste, reuse the component, and so on. Now my work is much more trying to create a good prompt, trying to explain the requirements in a clear, objective way and that maintains the project standards. That is, when I started, I usually got the standards that had to be followed very wrong. Now I have there a little cheat sheet file, I always ask it to refer to it, look, the standards are these, the guidelines are these, the best practices are these. So, when it creates a new component, it usually will try to follow these good practices. And not only that, as I can do it, a way of doing debugging also changed a lot. Ah, I found a bug. Okay. For some reason, a screen is loading wrong. Why is this screen loading wrong? So, first I reproduce the bug, collect some logs, copy them, throw them into the AI, say: X, Y, Z is happening. The steps I did were A, B and C and Z happened. Okay. From that, I ask it to create a document on why this may be happening, what code is involved in this. So, it generates a document for me, for example, I always ask it to generate a document, not simply answer me, because when you ask it to answer in the chat, it usually answers little. When you ask it to generate a document, it usually thinks more and gives you a more detailed answer. So, and usually, like, I will say not 100% of the time, but also not 90, 70% of the time, it discovers where the error is. It discovers, ah, the error is in the place, in location X. You go there to investigate and it really is there in location X. So, making a simple modification, sometimes, it even gives the alternative itself, it says: "Ah, if you do this, it fixes the bug." Or it says: "Look, I am not sure if this will fix the bug, but I have alternative A, B and C that you can try." Then I see which alternative, test the AI's alternatives, check whether it makes sense and if any of them makes sense and really fixes the bug, I ask it to implement it. I say: "Look, implement A, I will test it." So it implements A, I test it. Did it work? Did it not work? If it did not work, let's test B. I ask it to try B. I ask it to try C. And usually in this trial process of its, I also usually put some logs, I go seeing what is really happening. And it may be that in this process I discover that what the AI said does not make that much sense. Look, this fix of yours there, this correction of yours is not exactly the correction I needed. Or it may be that I discover that, oops, it is going down the right path, but such-and-such thing was missing. Then I explain, look, such-and-such thing was missing. So, it goes there and fixes it for me. So, a debugging process has also made things much easier for me. Yes, another thing that I find interesting is summarizing what was done in that story or in the bug that was fixed. Yes, because sometimes you fix a bug, you send it there to your peers to do code review, correct? But sometimes the person who will do the code review does not have the entire context of what was done. Sometimes he looks at the ticket, he sees, ah, it was such-and-such ticket. So, I know that the bug was this. But how was it fixed? What does this code that he changed modify in itself? So, I can also ask the AI to generate the PR description. So, it will generate the PR description for me saying, look, I changed this, this, this, because it changes this and this fixes the error. And it explains this nicely. So, usually, at the end of the correction process, I ask it to create a PR message, a PR description, which I will generate and so when someone does the code review they will already have a general context of why this fix really fixes the error and not simply have nothing in the description or I make a very, fix the error description. Understand? It makes it much easier.

Do you do code review with AI?
No, no. Code review I do manually. But we have a system there that when we submit the PR, first it will take, it is a system in two parts. When we submit the PR, like, we submit the PR description, there is a service there that takes our raw description and puts it into a standard and formats our description. After it formats our description into a standard, it runs an auto code review there that uses AI to do this. Yes, so we use this auto code review and we take a look at whether what the auto code review did makes sense, then we can accept it, sometimes it gets it very wrong. This one, this one is an AI that gets it very wrong, which is the auto code review one, that it does a code review and usually its review does not make sense. Sometimes it says: "Ah, why didn't you put the

non-null here? Since it is there, clearly it is there." Why didn't you check if it is not null? Since two lines above there is the check that it is not, that it cannot be null. So, this code review one gets it very wrong. But the description one is interesting.

And you commented about requirements, right, that you have to generate my better requirements, but isn't it the PO who passes these requirements to you or not?
So, we receive requirements there that come in the story, but sometimes the requirement is tangled among other documentations. So, the requirement is there, you have to make button x y z. Okay. But then there are the design guidelines. The design guideline is not in the ticket. There is another document. It is an external document. So, I need to see this external document. I have to copy and paste this part of this external document and put it in my prompt or ask it to generate a new document that includes not only the requirements, but the design guideline and also accessibility issues when I need it. So, I need to access an external document that is not directly in the story, it is not described in the story, but it is in an external document. So, because of this, I need to generate a new requirements document for myself.

And what difficulties do you find when using AI?
Ah, the biggest difficulty I think is having the patience to be able to describe the problem well. I think the biggest difficulty is this, it is being able to describe the problem that you want to attack. It is not simply you saying, you can say it in a very simple, very sloppy way and maybe the AI can do it. There is a great chance it will do it. But there is also a chance that your description will lack details, that you will forget things such as, for example, the project standards themselves and the AI doing that from its head. In its head, in quotation marks, right, in many quotation marks, doing that from its base. And sometimes we are in a hurry there, want to do something quickly, we do not have this patience to refine what we are going to say to the AI. I think this is one of the biggest difficulties. I started to improve this a little by using the AI's own chat to generate a better prompt for me. So I talk to it in a high-level way, super high-level even, using, not using so many standards and terms, and ask it from that to generate perhaps a prompt that I will use to feed itself. So, it generates a prompt for me, I ask it to do this in a separate document, usually it thinks more, and generates a prompt with better descriptions. With this I can reduce this difficulty a little, I will not have so, so much difficulty. Ah, but another thing that ends up happening to me a lot is the issue of the AI's context limit. Depending on how much code you make, you will fill your AI's context and it will start hallucinating a lot. I realized that usually there at 80 to 70% full context, it starts hallucinating a lot, it starts forgetting things. So, this is a difficulty for very large tasks. A very large task, it ends up getting very lost. So, we have to think when giving the descriptions to it, of making it divide the tasks into smaller and smaller activities. Because with smaller activities, it will have a better context limit to work with and it will not hallucinate so much. So, with small tasks, I can put things in its context such as the standards, I can put things in its context, such as, for example, architectural definitions, that if I leave a very large task and simply ask it to do everything at once, it will forget the standards, it will forget the architectural decisions and will start doing things from its, from its head and it will not work, it will be no good. It will not be something cool, it will not turn out good. So this context limit that we can work with is a difficulty. So usually we try to get around this by creating new documents. So yes, there are many, many stages that include us creating documents that will serve only to feed the AI at that time when we are doing the activity. So, it will be a document that it will take, oops, the general context is this, the progress we are making is this. We will divide this activity into 1 2 3 4 5 6 7 8 stages, if necessary, and each small stage of this we will have to have well described. And from this, what do we have to do? We have to open a new chat and ask it to do only that little thing, validate, open a new chat, ask it to do only that little thing, precisely so as not to fill the AI's context. This also happens a lot during bug investigation, we start telling it: "Ah, see such-and-such thing, see such-and-such thing, see such-and-such thing." When we see it, its context is full and it starts to hallucinate and no longer can really find the problem,

solve the problem, it really starts hallucinating and starts generating trash code and starts doing things it should not. So this context issue is the biggest difficulty there is. Managing the context.

For example, couldn't you pass a description, then you saw that something was missing, do one more interaction saying what was missing. You could do that, right?
You could do that, but then it enters the context part. You asking, when you ask it to redo something, it already has stored in its memory the whole previous context of what it did and more this new information that you added. So, either it pollutes, sometimes it does a lot. it pollutes the document. You have a nice specification document and it simply goes there and pollutes the document saying: "Oops, it is not to do it not to do it like this, it is to do it like this." And this makes it, this small instruction of do not do it like this and do it like this, already harms it when it, effectively, programs. I understand. That it keeps thinking all the time: "Oops, I do not have to do this, I have to do it this way." And sometimes it extrapolates. It hallucinates and extrapolates. Oops, if it cannot be done this way, I will modify this here that had nothing to do with my original request. So, the iterative process with it, especially when generating a document, is a problem. What is cool to do is, first you talk to it only in chat mode there. Just talk to it in chat mode. Ah, you take the good part and discard what is not, you open another chat, because you clean the context, it will start from zero. So, many times we have to do this, to avoid polluting the context, polluting with useless things.

And how was your process to start using AI? Did you find it difficult, easy?
At the beginning I found it a little complicated, mainly the issue of not knowing how to ask the AI in the correct way, this issue of generating a document, before starting. I think I made every beginner's mistake which is just arriving at the AI and saying: "Do this." You have to make this green button. Make the green button for me." Then the AI will try to make the green button. You did not give a specification, you did not say what the size of the button was, you did not say what needed to be there, how it needed to be, you did not say whether it needed to reuse some component, you only said: "Do such-and-such thing." It will try to do it its own way, it will have the context of your codebase. It will look in each place: "Oops, I found this, this, I will do it this way." And maybe, and at the beginning it even worked, but as the difficulty of the activity grows, maybe you simply asking it to do something in such a broad way, was not working so well. So, for this, we started developing techniques so that we could optimize how the AI was doing things and so that it would make fewer and fewer mistakes.

And how do you feel using AI today? Do you prefer working with it or as it was before?
I will say with AI, mainly because a good part of a Dev's work is not writing code, it is thinking about how to solve the problem. The tedious part, which is only you writing code, is only the final part, it is as if you were there, as if you were a bricklayer and the whole part of thinking about how it will be measured, what the measurements will be, that is the complicated part. The simple part is you laying the bricks, correct? But a very tiring part. It is simple in the sense that it is something repetitive. You know how to do it. It is repetitive, you will spend there two, three, four hours just laying bricks. So, coding manually is that. It is you typing. Typing the code. You know where you want to get, but you will have to go through this. With AI, it is much simpler for me to ask it to generate this code again, this new code for me. Especially if you need to generate new code. Because in the end, what changes is that before I would copy and paste another code that would be ready. I copy Ah, I want to create a new class, copy and paste, create a new file, copy and paste what I needed and make the modifications that I need. That is what we did before. Now I can pass this work to the AI, we have to keep generating code, keep writing the code for me. Of course, sooner or later I will have to go there and touch it because maybe it is faster than simply modifying, maybe, it covers a variable, or maybe it is faster for me to modify a color, maybe it is faster to modify a pattern, manually. Okay? But the part of generating something new, the part of

having to dive into the code, investigate what is going wrong, why such-and-such thing is failing, I can delegate this to the AI. And it makes my life much easier, especially in a very large codebase.

And are there situations in which AI does not help as expected?
Yes, yes. Ah, although it helps a lot in bug correction, there are things it does not do. Yes, hum, ah, let's suppose, I need to reproduce a bug. The AI does not reproduce the bug for me. Sometimes, I mainly, let me see. It does not do my work in the manual, which is manual testing, for example. Ah, and after it validates, that it does its things, I have no way to make it, at least not yet. I do not know if it is possible to do this. It cannot check on the device for me. So, it made the code, okay? Now I need to test. I make the build, send it to my device and now I have to do the manual part which is me checking whether everything is right, whether it is working, whether the button is clicking, whether it is in the right position, whether it is where it should be, whether my adding this component did not break anything. So, all of this is a manual verification work that I have to do manually. So, things that involve the physical world are still my work, as a developer.

And do you have any practice or guidance, about what could improve in the use of AI? What do you think is still missing to use it better in development?
On my part, I think there is still a matter of my knowledge too, I think learning a little more about how to use these tools. I think I use it at an OK level, but not yet at a super level. I think it is more a matter of mine. I think there is indeed a lack of training on how to use it better, but we also have to consider that it is a relatively new area, right? So, it is a new area and a bit hands-on. So, you learn kind of the hard way. Like, you learn, practically, as we learned to program, which was by trial and error. So, like any new skill, we need to try by trial and error. But I miss a formal training, like, what are the best practices, how it works. I end up having to watch tutorials on YouTube, but even so, these tutorials are very vague and depending on the area for which you are programming, the content is less comprehensive. For example, there is a lot of content on the internet about how to make a website with AI, something that you open on YouTube and ask how to make a website with AI and it will explain to you how to do it, it will tell you the prompts you have to make and usually it will be there with the cake recipe that is usually the Veracel combo, right, which is Next JS with Node, some service like that. So, there is already a kind of half-ready cake recipe there on how to do this. But depending on the technology you are working with, there is not such an easy cake recipe on YouTube, for example. Especially a relatively niche technology, such as, for example, Android and more specifically Android TV. So, yes, there is a bit of this type of content missing on the internet. So, we have to end up taking something that is more general, usually TypeScript, JavaScript, some Java and try to apply it to another reality.